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# Analysis of Frequency Domain Spectroscopy Characteristics and Machine Learning-Based Classification of Transformer Insulation Condition

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EE576 – High Voltage Engineering — Mini Project Report

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## 0. Introduction

Power transformers are among the most critical assets in electrical power transmission and distribution systems. The condition of the cellulose-based insulation system — consisting of mineral oil, paper, and pressboard — directly governs transformer reliability and service lifetime [1]. Moisture accumulation within the cellulose structure significantly reduces dielectric strength, accelerates insulation ageing, and may ultimately lead to catastrophic transformer failure.

Frequency Domain Dielectric Spectroscopy (FDS) is a widely accepted non-destructive diagnostic technique for evaluating transformer insulation condition [1]. In FDS, the dielectric loss tangent  $\tan \delta$  is measured over a wide frequency range (typically 0.1 mHz to 1 kHz). However, the measured response is influenced simultaneously by moisture content, temperature, oil conductivity, and insulation geometry, which makes direct visual interpretation unreliable.

Machine learning (ML) methods have been demonstrated to be effective for extracting moisture-related patterns from FDS datasets and automating insulation condition assessment [2]. This report applies Principal Component Analysis (PCA) for dimensionality reduction and trains Support Vector Machine (SVM) [3] and neural-network [4] classifiers on the Group 03 FDS dataset. SMOTE-based data augmentation [5] is further explored to address the limited dataset size.

The insulation condition is classified into three classes:

$$\text{Dry } (MC < 1\%) \quad \text{Intermediate } (1\% \leq MC \leq 3\%) \quad \text{Wet } (MC > 3\%)$$

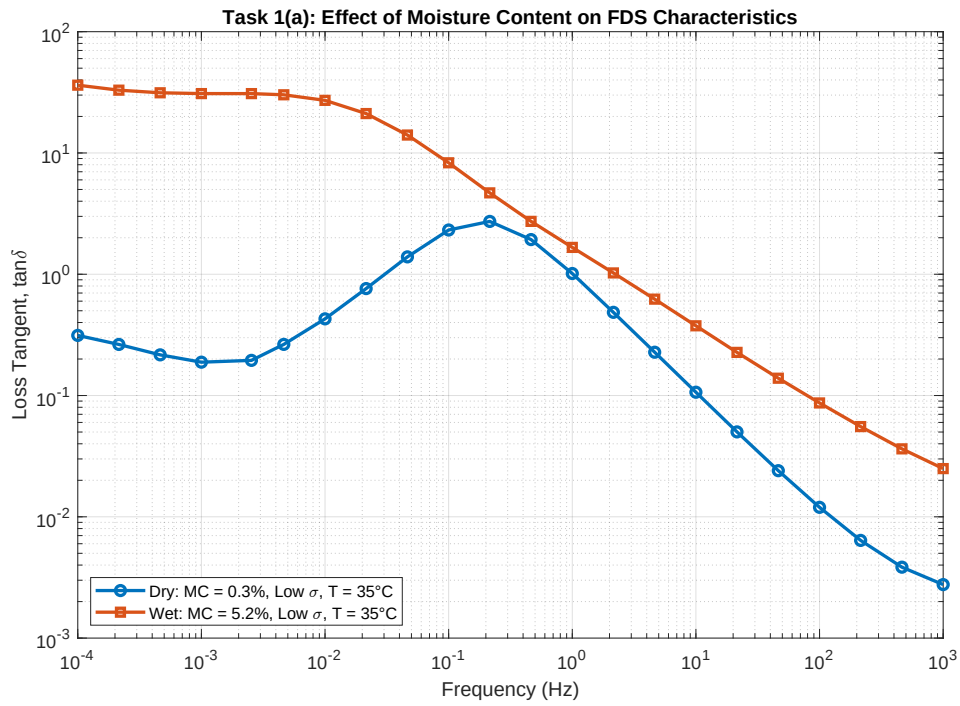
The overall workflow is:

$$\text{FDS data} \rightarrow \text{FDS analysis} \rightarrow \text{PCA} \rightarrow \text{ML classification} \rightarrow \text{Performance evaluation}$$

## 1. Task 01: Analysis of FDS Characteristics

### 1.1 Effect of Moisture Content

Dry ( $MC = 0.3\%$ ) and wet ( $MC = 5.2\%$ ) insulation samples were compared at low oil conductivity and  $T = 35^\circ\text{C}$  [2].

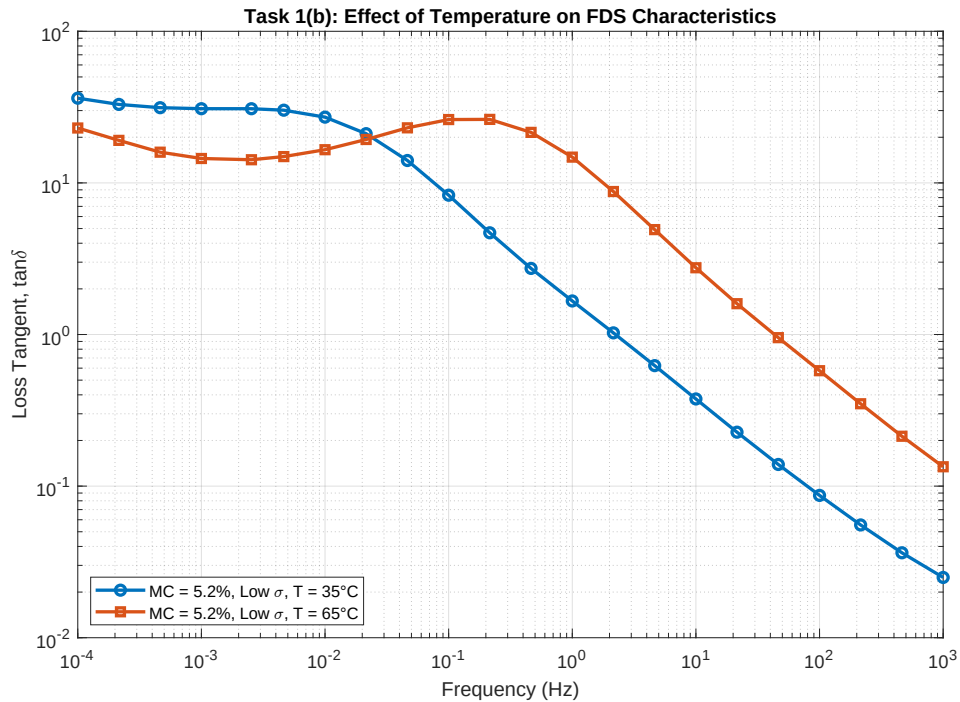


**Figure 1:** Effect of moisture content on FDS loss tangent response.

From Fig. 1, the wet sample exhibits a significantly higher loss tangent, especially at low frequencies. Moisture increases polarisation and ionic conduction within the cellulose structure, making the low-frequency FDS region the most diagnostic for moisture estimation [1].

## 1.2 Effect of Temperature

FDS responses at  $T = 35^\circ\text{C}$  and  $T = 65^\circ\text{C}$  were compared at  $MC = 5.2\%$  with low oil conductivity.

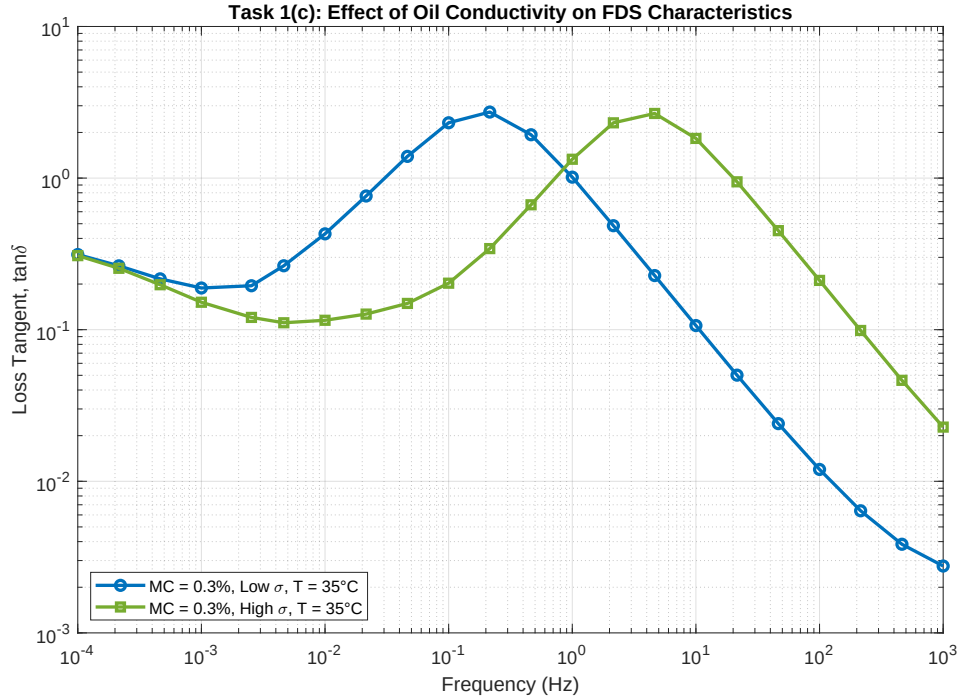


**Figure 2:** Effect of temperature on FDS loss tangent response.

Higher temperature accelerates polarisation and charge transport processes, shifting the dielectric response curve across the frequency axis. The same insulation sample therefore produces a different FDS curve at different temperatures, introducing ambiguity in moisture estimation [1].

### 1.3 Effect of Oil Conductivity

Low and high oil conductivity conditions were compared at  $MC = 0.3\%$  and  $T = 35^\circ\text{C}$ .



**Figure 3:** Effect of oil conductivity on FDS loss tangent response.

High oil conductivity raises the loss tangent in a manner that can be confused with moisture-related changes, emphasising the need for multi-factor analysis before drawing conclusions about insulation condition.

### 1.4 Discussion: Challenges of Direct Moisture Estimation

The measured FDS response is a function of multiple parameters simultaneously [1]:

$$\tan \delta = F(MC, T, \sigma_{\text{oil}}, f)$$

where  $MC$  is moisture content,  $T$  is temperature,  $\sigma_{\text{oil}}$  is oil conductivity, and  $f$  is frequency. As demonstrated in Sections 1–3, all three physical parameters produce qualitatively similar shifts in the loss tangent curve, particularly a rise at low frequencies. Direct visual interpretation is therefore unreliable. Machine learning methods address this by learning the multivariate pattern across all 22 frequency points simultaneously [2].

## 2. Task 02: Feature Correlation Analysis and PCA

### 2.1 Motivation for Dimensionality Reduction

Adjacent frequency points in an FDS curve tend to carry very similar information because the loss tangent varies smoothly with frequency. If many input features are highly correlated, training a classifier on all 22 raw features may introduce redundancy, increase training complexity, and reduce generalisation ability. A Pearson correlation analysis was therefore performed before applying PCA.

## 2.2 Correlation Analysis

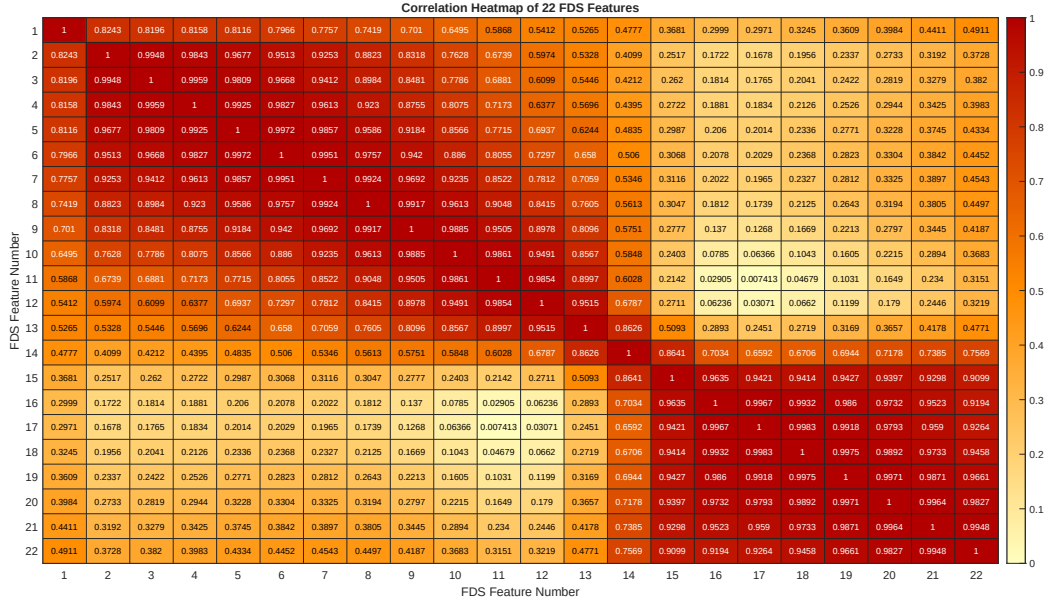


Figure 4: Correlation heatmap of the 22 FDS features.

Table 1: Summary of feature correlation analysis.

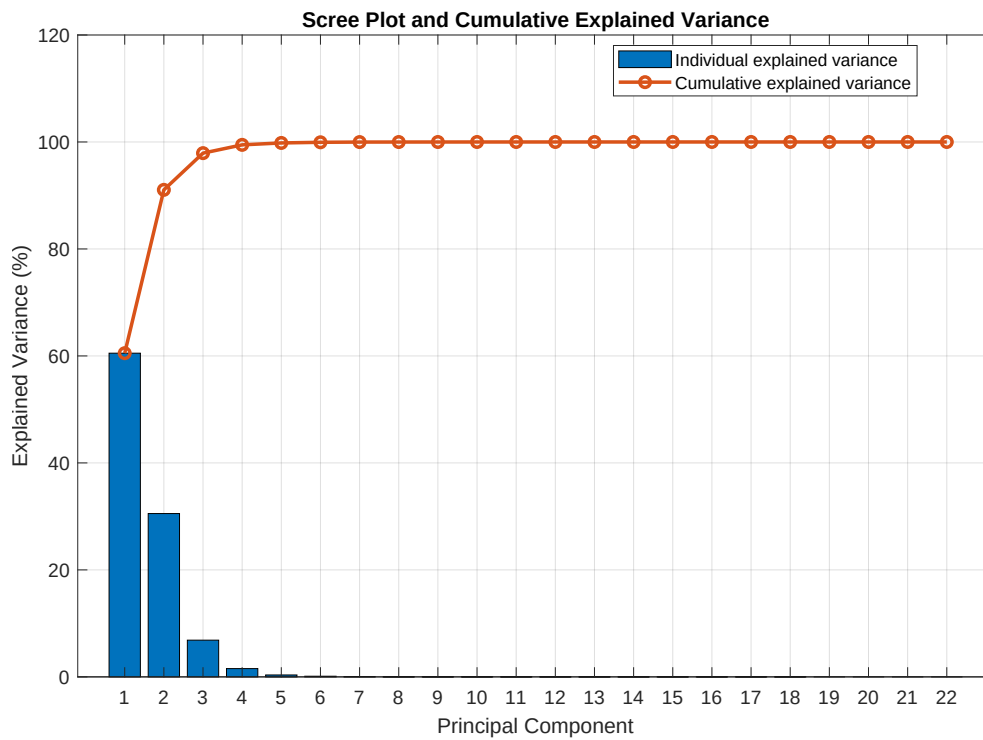
| Statistic                         | Value      |
|-----------------------------------|------------|
| Total feature pairs               | 231        |
| Pairs with $ r  > 0.80$           | 81 (35.1%) |
| Pairs with $ r  > 0.90$           | 60 (26.0%) |
| Pairs with $ r  > 0.95$           | 43 (18.6%) |
| Mean adjacent-feature correlation | 0.9698     |

With an average adjacent-feature correlation of 0.9698, the raw 22-feature set contains a high degree of redundancy, motivating dimensionality reduction via PCA.

## 2.3 Principal Component Analysis

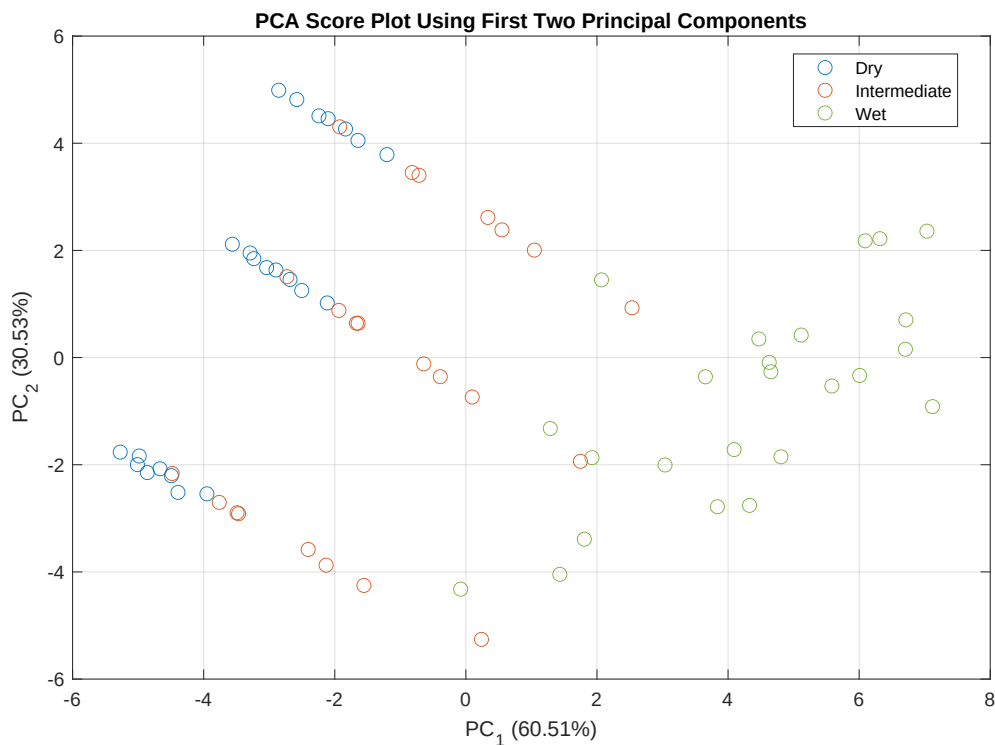
PCA was applied after  $\log_{10}$ -transformation and standardisation of the features [4]. The  $\log_{10}$  transform is necessary because loss tangent values span several decades; standardisation ensures unit variance across features before PCA:

$$z = \frac{x - \mu}{\sigma}, \quad [f_1, \dots, f_{22}] \xrightarrow{\text{PCA}} [PC_1, \dots, PC_{22}]$$



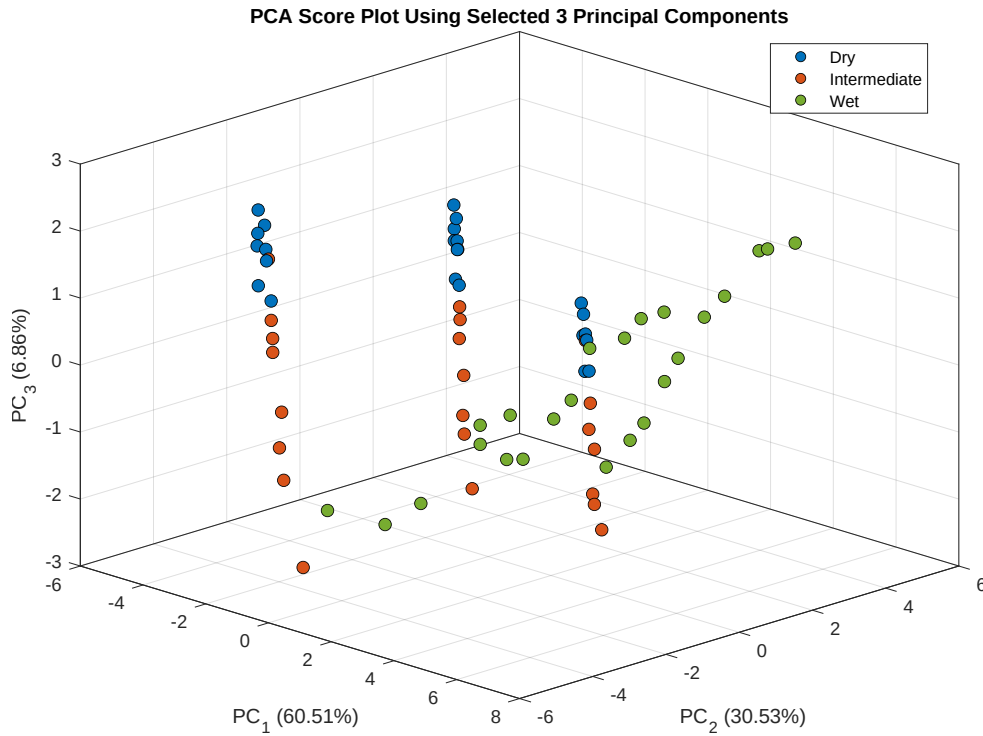
**Figure 5:** Scree plot and cumulative explained variance of principal components.

$PC_1$  explains **60.51%** of total variance;  $PC_2$  explains a further **30.53%**. Together,  $PC_1 + PC_2$  capture **91.04%** of total variance. Including  $PC_3$  raises this to **97.90%**.



**Figure 6:** PCA score plot: first two principal components showing clear class separation [2].

Fig. 6 shows clear visual separation of the three insulation classes in the  $PC_1$ – $PC_2$  plane, consistent with the findings of Wijethunga et al. [2]. A three-dimensional view including  $PC_3$  is shown in Fig. 7.



**Figure 7:** PCA score plot: first three principal components ( $PC_1$ – $PC_3$ ).

For classification in Task 03, all 22 standardised features were retained in addition to the 2-PC representation, since projecting onto two components discards the remaining 8.96% of variance which may carry class-discriminating information.

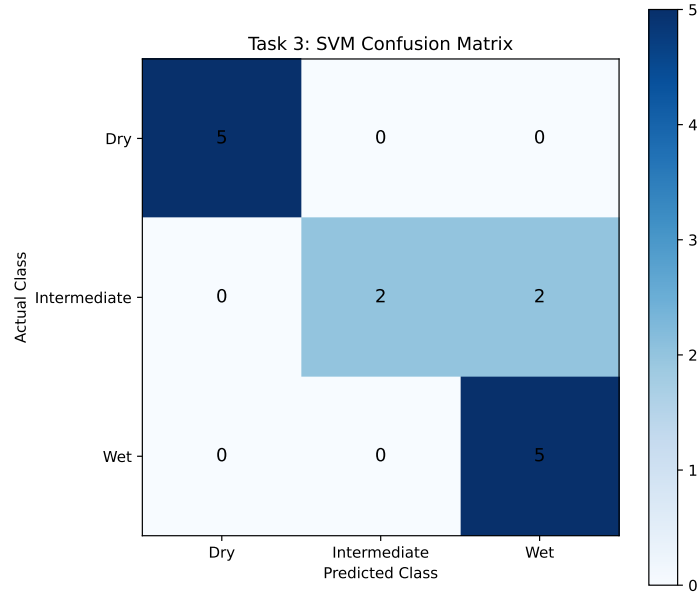
### 3. Task 03: Machine Learning Classification

#### 3.1 Dataset Preparation

The Group 03 dataset contains 70 samples (23 Dry, 23 Intermediate, 24 Wet) across 22 FDS frequency features. An 80/20 stratified split yields 56 training samples and 14 test samples. All features were  $\log_{10}$ -transformed and standardised using training-set statistics only, ensuring no data leakage into the test set.

#### 3.2 2PC-SVM Classifier

Following the methodology of Wijethunga et al. [2], a linear Support Vector Machine [3] was first trained on  $PC_1$  and  $PC_2$  only (training-set PCA:  $PC_1 = 62.04\%$ ,  $PC_2 = 29.31\%$ , total = 91.36%). The SVM finds a hyperplane  $g(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b = 0$  that maximally separates the three classes via a one-versus-one strategy [3].



**Figure 8:** Confusion matrix – 2PC-SVM classifier (test accuracy 85.71%).

**Table 2:** Classification report – 2PC-SVM.

| Class           | Precision   | Recall | $F_1$ | Support |
|-----------------|-------------|--------|-------|---------|
| Dry             | 1.00        | 1.00   | 1.00  | 5       |
| Intermediate    | 1.00        | 0.50   | 0.67  | 4       |
| Wet             | 0.71        | 1.00   | 0.83  | 5       |
| <b>Accuracy</b> | <b>0.86</b> |        |       | 14      |

Two intermediate samples were misclassified as wet, reflecting class overlap in the reduced 2D PCA space.

## 4. Task 04: Performance Evaluation and Further Analysis

### 4.1 Evaluation Metrics

Models were evaluated using accuracy, precision, recall,  $F_1$ -score, and confusion matrix [4]:

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}}, \quad F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

### 4.2 Tuned All-Feature SVM

An SVM [3] trained on all 22 standardised features with grid-search hyperparameter tuning (linear kernel,  $C = 100$ , five-fold cross-validation) achieved a cross-validation accuracy of 96.52%.

**Table 3:** Classification report – tuned all-feature SVM.

| Class           | Precision   | Recall | $F_1$ | Support |
|-----------------|-------------|--------|-------|---------|
| Dry             | 1.00        | 1.00   | 1.00  | 5       |
| Intermediate    | 1.00        | 0.75   | 0.86  | 4       |
| Wet             | 0.83        | 1.00   | 0.91  | 5       |
| <b>Accuracy</b> | <b>0.93</b> |        |       | 14      |

Testing accuracy improved from **85.71%** (2PC-SVM) to **92.86%** (tuned all-feature SVM), corresponding to 13 out of 14 correct classifications.

## 5. Further analyses

### 5.1 Neural Network Analysis

A compact feedforward neural network [4] with architecture  $22 \rightarrow 16 \rightarrow 8 \rightarrow 3$  (ReLU activations, softmax output, dropout and  $L_2$  regularisation, early stopping) was trained on all 22 standardised features as a further analysis.

**Table 4:** Classification report – basic neural network.

| Class           | Precision   | Recall | $F_1$ | Support |
|-----------------|-------------|--------|-------|---------|
| Dry             | 1.00        | 0.80   | 0.89  | 5       |
| Intermediate    | 0.75        | 0.75   | 0.75  | 4       |
| Wet             | 0.83        | 1.00   | 0.91  | 5       |
| <b>Accuracy</b> | <b>0.86</b> |        |       | 14      |

The basic neural network matched but did not exceed the 2PC-SVM. As discussed in [4], deep networks generally require large training sets to generalise reliably; with only 70 samples, the SVM is better suited to this problem.

### 5.2 SMOTE-Augmented Neural Network

To address the small training set, Synthetic Minority Over-sampling Technique (SMOTE) [5] was applied *only* to the 56 training samples, expanding the training set from 56 to 280 synthetic samples (93 Dry, 93 Intermediate, 94 Wet). SMOTE generates new samples by linear interpolation between existing minority-class instances in feature space [5]. The final test evaluation was performed exclusively on the original 14 real samples to ensure a fair comparison.

A larger network ( $22 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 3$ , 1427 parameters) was then trained on the augmented set.

**Table 5:** Classification report – SMOTE-based neural network.

| Class           | Precision   | Recall | $F_1$ | Support |
|-----------------|-------------|--------|-------|---------|
| Dry             | 1.00        | 1.00   | 1.00  | 5       |
| Intermediate    | 1.00        | 0.75   | 0.86  | 4       |
| Wet             | 0.83        | 1.00   | 0.91  | 5       |
| <b>Accuracy</b> | <b>0.93</b> |        |       | 14      |

SMOTE augmentation improved neural network test accuracy from **85.71%** to **92.86%**, matching the tuned all-feature SVM. Final test accuracy:  $13/14 = 92.86\%$  on the original real test set.

### 5.3 Overall Model Comparison

**Table 6:** Summary comparison of all developed models.

| Model                     | Train / Val Acc. | Test Acc.    | Key observation                                                 |
|---------------------------|------------------|--------------|-----------------------------------------------------------------|
| 2PC-SVM [3, 2]            | 89.3%            | 85.7%        | Useful for visualisation; 2 intermediate samples misclassified. |
| Tuned all-feature SVM [3] | 100.0%           | <b>92.9%</b> | Best SVM; selected as main classifier.                          |
| Basic neural network [4]  | 93.2 / 91.7%     | 85.7%        | Insufficient training data to outperform SVM.                   |
| SMOTE-NN [5, 4]           | 100.0 / 100.0%   | <b>92.9%</b> | Data augmentation matches tuned SVM performance.                |

The tuned all-feature SVM is selected as the primary classifier: it achieves the highest test accuracy, requires no synthetic data, and is computationally lightweight relative to neural networks.

### 5.4 Suggestions for Further Improvement

1. **Larger and more diverse dataset.** A wider range of moisture levels, temperatures, oil conductivities, and ageing stages would improve generalisation, as noted by Wijethunga et al. [2].
2. **Temperature and oil conductivity compensation.** Pre-processing to normalise the FDS curve for known temperature and conductivity [1] would reduce inter-factor interference before classification.
3. **Improved feature reduction.** Comparing different numbers of PCA components, or using supervised feature selection, may identify a more compact and informative feature subset [4].
4. **Additional classification algorithms.** Random Forest, Gradient Boosting,  $k$ -NN, and ensemble methods should be benchmarked against SVM [3] using identical train/test splits.
5. **Higher-quality synthetic data.** SMOTE [5] generates samples by linear interpolation; generative adversarial networks or variational autoencoders [4] could produce more realistic augmented FDS data.

## 6. Conclusion

FDS analysis confirmed that transformer pressboard insulation response is driven simultaneously by moisture content, temperature, and oil conductivity [1]. Moisture raises the loss tangent predominantly at low frequencies; temperature shifts the entire response curve; and oil conductivity produces changes that can mimic moisture effects. These overlapping influences make direct visual interpretation unreliable, motivating the use of machine learning [2].

Correlation analysis showed that 43 out of 231 feature pairs have  $|r| > 0.95$  (mean adjacent-feature correlation 0.9698), confirming high redundancy in the raw 22-feature set. PCA condensed this to two principal components capturing **91.04%** of total variance, providing clear three-class visualisation consistent with [2].

**Final result:** Tuned all-feature SVM [3] and SMOTE-NN [5, 4] both achieve **92.86%** test accuracy (13/14 correct). The tuned SVM is selected as the primary classifier due to its simplicity and suitability for small, high-dimensional datasets.

The combination of FDS analysis [1], PCA visualisation, SVM classification [3], and SMOTE-based augmentation [5] provides a practical and effective approach to automated transformer insulation

condition assessment, building on the methodology first demonstrated by Wijethunga et al. [2].

## References

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